Translating Italian University Strategies into Funding Opportunity

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Technical Report

Team Members
Filippa Grönberg (Delegate) —gronberg.filippsofi@studenti.luiss.it
Luka Burmazevic — luka.burmazevic@studenti.luiss.it
Evelina Ristovska — evelina.ristovska@studenti.luiss.it

Company: EY
Academic Year 2025/2026

Section 1: Introduction

Italian universities are required to produce multi-year Strategic Plans that define their priorities across research, education, internationalization, and governance. At the same time, a large and constantly evolving landscape of competitive funding calls spans the Italian National Recovery and Resilience Plan (PNRR) and the European Union's Horizon Europe programme, specifically the WIDERA (Widening Participation and Spreading Excellence) strand [6], offering substantial resources for academic institutions that can demonstrate alignment with call objectives.

The challenge is twofold: strategic plans are lengthy documents (often 100+ pages, written in Italian), while funding calls are highly technical and domain-specific. Manually identifying which calls are most relevant to a given university's strategic priorities requires considerable expert effort and is difficult to scale across multiple institutions simultaneously.

1.1 Case Study and Scope

This project addresses the above challenge for six major Italian universities, selected to represent a balanced cross-section..." Remove the first sentence and keep only the second one after the bullet list, or better yet move the cross-section sentence to before the bullet list and remove the duplicate entirely.

- **Bocconi University (Piano Strategico 2021–2025 & Vision 2030)**
- **LUISS Guido Carli (Piano Strategico 2021–2024)**
- **LUM University (Piano Strategico di Ateneo 2021–2025)**
- **Sapienza Università di Roma (Piano Strategico 2021–2027)**
- **Politecnico di Milano (Strategic Plan 2023–2025)**
- **Università degli Studi di Napoli Federico II (Piano Strategico 2021–2026)**

1.2 Goal

The goal of the project is to build an automated recommendation system that, given a university's strategic plan as input, returns the top-3 most relevant funding calls, ranked by alignment score, accompanied by:

- **Supporting evidence based on matched strategic themes**
- **Identified gaps between the university's priorities and the call's requirements**
- **Confidence-graded executive summaries suitable for stakeholder communication**

1.3 Approach Overview

The solution is a three-phase Retrieval-Augmented Generation (RAG) pipeline, an architecture designed to ground language model outputs in external evidence, directly addressing the hallucination and stale knowledge limitations inherent to standard LLMs.

Phase 1 (the baseline) performs dense semantic retrieval and multi-component score aggregation to produce a ranked call list.

Phase 2 adds a deterministic explainability layer using a bilingual keyword lexicon.

Phase 3 uses a locally-run LLM (Qwen3:0.6b via Ollama) to generate grounded stakeholder briefings. Source documents are in Italian and English.

Section 2: Methods

2.1 Data Collection

All data consists of PDF documents – no web scraping or external APIs were used. The six strategic plans were downloaded from official university websites and are primarily in Italian, with some mixed

English/Italian content (Politecnico di Milano). Five PNRR funding calls were collected from the official MUR portal, covering Partenariati Estesi [7], Ecosistemi dell'Innovazione [8], Centri Nazionali [9], PRIN 2022 [10], and PRIN PNRR 2022 [11]. Additionally, 55 EU Horizon Europe WIDERA call summaries were retrieved from the European Commission website [6]. Strategic plans range from approximately 50 to 200 pages; PNRR funding call documents range from 20 to 50 pages, while WIDERA summaries are considerably shorter at 1 to 3 pages, a difference that has implications for retrieval performance discussed in Section 3.1.

2.2 Pre-processing and Chunking

Text was extracted page-by-page using pypdf, with very short pages discarded as noise. A cleaning step removed null bytes and collapsed all whitespace. Both document types were split into overlapping chunks using a sliding-window approach, consistent with standard RAG ingestion practice where chunking granularity directly affects retrieval relevance. Funding call documents were chunked at a larger size to preserve the broader context of call requirements, while strategic plan chunks were kept smaller so that each acts as a focused, fine-grained query during retrieval.

2.3 Pipeline Architecture

The system is implemented across two notebooks; ingest.ipynb for building the vector store and query.ipynb for the full recommendation flow, plus supporting files (llm_explainer.py, pipeline_core_methods.py) and app.py which launches the Streamlit interface. Table 1 summarizes the four-phase pipeline, from corpus ingestion to stakeholder briefing generation.

Phase	Input	Process	Output
Phase 0: Ingest	Funding call PDFs	Extract → clean → chunk (1800 chars, overlap 250) → embed	ChromaDB vector store
Phase 1: Rank	Strategic plan PDFs	Chunk SP (1200 chars) → query ChromaDBtop-30 → aggregate scores	Top-3 ranked calls per university
Phase 2: XAI	Phase 1 results	Theme lexicon matching + gap detection	Matched themes + improvement actions
Phase 3: LLM	Phase 2 XAI objects	Qwen3:0.6b prompt → parse JSON + plain text	Stakeholder briefing with citations + confidence

Table 1*. Four-phase pipeline architecture: inputs, processes, and outputs*

2.4 Models

Embedding Model

All text, both funding call chunks and strategic plan chunks are embedded using Qwen/Qwen3-Embedding-0.6B [1] via the SentenceTransformers library. We selected this model because it is explicitly multilingual and performs well on Italian-language text, which is critical given that all strategic plans are written in Italian and the funding calls are predominantly in English. A single shared model ensures that SP and call embeddings occupy the same semantic vector space, enabling meaningful cosine similarity comparisons.

Vector Store

ChromaDB [4] is used as the persistent vector store. Funding call chunks are upserted in batches of 128 with their embeddings and metadata (source file, page number, chunk index). During query time, each strategic plan (SP) chunk is used as a vector query to retrieve the top 30 most similar call chunks. Retrieval operates via dense vector search using cosine similarity between query and document embeddings, enabling semantic matching even when the SP text and funding call use different terminology for the same concept. This is a key advantage of dense retrieval over sparse approaches such as BM25, and is particularly important here given that strategic plans are written in Italian while funding calls are predominantly in English.

2.5 Hyperparameter Configuration

The parameters used in the project were set empirically by inspecting intermediate retrieval and generation outputs across multiple universities. The configuration covers three areas:

Retrieval and ranking

N_RESULTS = 30: controls the number of candidate chunks retrieved per SP query chunk, set high to maximise recall before re-ranking.

STRONG_HIT_THRESHOLD = 0.55: is the minimum cosine similarity for a retrieved chunk to count as a strong match, which feeds directly into the coverage score computation described in Section 2.6.

TOP_K_CALLS = 3: sets the number of final recommendations per university, and EVIDENCE_PER_CALL = 3: limits the supporting chunks surfaced per call in the output.

Gap detection

GAP_MIN_CALL_SIGNAL = 0.20 & GAP_MAX_SP_SIGNAL = 0.10: work as a pair — a theme is flagged as a key gap only when the call scores above 0.20 on that theme while the strategic plan scores below 0.10, meaning the call strongly emphasises an area the university has not yet addressed.

XAI_TOP_THEMES = 4 & XAI_MAX_GAPS = 4: cap the number of matched themes and gaps reported per call respectively.

LLM generation

PHASE3_TEMPERATURE = 0.1: a low temperature yields near-deterministic results. Running the same prompt twice yields near-identical text, which is a deliberate choice: reproducibility and evidence fidelity take priority.

DEFAULT_NUM_PREDICT = 2000: caps the maximum tokens generated per response; if the initial response fails to parse, typically due to truncation — the system retries with a doubled token generation limit before falling back to a reformat request.

DEFAULT_NUM_CTX = 8192: sets the context window passed to Ollama. The /no_think prefix in the prompt is a Qwen3-specific instruction that suppresses internal chain-of-thought reasoning, ensuring the model returns structured output directly.

PHASE3_MIN_CITATIONS_PER_CALL = 2: enforces that each recommended call must be backed by at least two source citations, validated programmatically before output is accepted.

2.6 Scoring, Evaluation, and Baseline

Calls are ranked using three complementary score components, each designed to capture a different dimension of alignment. The semantic score is the average cosine similarity of the top retrieved chunks for a given call, measuring direct textual alignment. The coverage score counts the number of distinct pages of the call document that were retrieved above the strong hit threshold, normalized to a maximum of six pages, rewarding calls that are matched across multiple sections rather than a single passage. The consistency score counts the number of retrieved chunks exceeding the strong hit threshold, normalized to five, rewarding calls with repeatedly high similarity rather than isolated peaks. These three components are combined into a weighted final score (semantic: 0.7, coverage: 0.2, consistency: 0.1). Semantic similarity is assigned the highest weight as it most directly captures thematic alignment between SP priorities and call requirements. Coverage is weighted second to reward calls that match broadly across the call document. Consistency receives the lowest weight as it serves primarily as a tiebreaker to filter out spurious high-similarity matches.

Since no ground-truth dataset of correct university–call pairings exists, evaluation is qualitative, based on expert review of the top recommendations. The baseline is Phase 1 alone: pure semantic retrieval with no explanations. The full system (Phases 1+2+3) is evaluated against it qualitatively: the baseline correctly identifies relevant calls but offers no interpretability. The full pipeline adds structured, evidence-backed explanations and confidence ratings that make recommendations usable by non-technical stakeholders such as university grant offices.

2.7 Explainability Layer (Phase 2)

To make recommendations transparent and actionable, Phase 2 adds a deterministic XAI (Explainable AI) layer on top of the ranked results. A bilingual theme lexicon maps eight strategic categories to Italian and English keyword lists, with between 14 and 19 terms per theme. Keywords were selected to cover the dominant terminology in both PNRR funding call documents and Italian university strategic plans. Matching uses exact substring counting on normalized lowercase text, meaning multi-word phrases (e.g. 'technology transfer', 'trasferimento tecnologico') are preferred over single tokens to reduce false matches.

Theme	Sample Keywords (EN / IT)
AI, Data & Digital Transformation	artificial intelligence, machine learning, big data, digital transition / intelligenza artificiale, apprendimento automatico, dati, transizione digitale
Internationalization & Collaboration	consortium, mobility, cross-border, partnership / consorzio, mobilità, partenariato, collaborazione

Table 2. *Examples of themes with matched keywords*

Table 2 illustrates an example of theme lexicon keywords – the entire lexicon covers the following themes:

- Sustainability & Climate
- AI, Data & Digital Transformation
- Internationalization & Collaboration
- Innovation Transfer & Industry
- Skills, Education & Capacity Building
- Inclusion, Gender & Social Impact
- Research Infrastructure & Excellence
- Governance, Policy & Reform

For each recommended call, the system computes a match strength per theme by comparing keyword frequency in the call evidence against the SP text. Themes where the call scores above 0.20 but the SP scores below 0.10 are flagged as key gaps, and a concrete improvement action is generated for each, guiding the university on how to strengthen its strategic plan to better align with that funding call.

2.8 LLM Output Validation (Phase 3)

The language model used in Phase 3 is Qwen3:0.6b [2], served locally via Ollama [3]. The model receives a structured JSON payload containing the top-3 ranked calls with their XAI objects: matched themes, key gaps, and supporting chunks with citations, alongside a detailed instruction prompt. It is required to return two outputs: a structured JSON block and a plain-text stakeholder briefing. Every recommended call must include at least two explicit source citations (source file and page number) drawn from the supporting evidence chunks. A temperature of 0.1 was deliberately set low to keep outputs near-deterministic and tightly grounded in the provided evidence. In a task like this, where recommendations must be justifiable and traceable, consistency and accuracy to the source material matter more than varied or creative phrasing, with the LLM handling reasoning and synthesis on top of what the retrieval layer already established.

Following generation, an automated post-processing step validates citation coverage for each recommended call. If the model failed to include citations that were present in the supporting evidence, the system automatically injects them from the retrieved chunks. If a call still falls below the minimum citation threshold after this correction, its confidence rating is downgraded to Low, ensuring the pipeline always produces output while transparently signalling weaker evidence through the confidence field.

2.9 User Interface

A Streamlit [5] web application was developed as the final delivery surface for the system. Streamlit was chosen because it enables a functional, stakeholder-facing interface to be built directly on top of existing Python modules with minimal frontend overhead, which was appropriate given that web development was not the focus of this project. The interface allows users to either select one of the six pre-processed university strategic plans from a sidebar dropdown or upload a new strategic plan PDF directly. Upon upload, the full three-phase pipeline is executed in real time – text extraction, chunking, semantic retrieval, XAI scoring, and LLM briefing generation. Key pipeline parameters such as chunk size, retrieval depth, scoring thresholds, and the number of top calls returned are exposed as configurable inputs, allowing users to explore how different settings affect the ranking of funding calls, with predefined defaults available for immediate use. Results are displayed interactively across expandable sections covering matched themes, key gaps, supporting evidence with page-level citations, and the executive briefing with confidence ratings. Pre-computed results for the six demo universities are loaded from saved JSON outputs, allowing instant exploration without re-running the pipeline. It should be noted that the real-time upload path requires a locally running Ollama instance with the qwen3:0.6b model pulled, as well as the project virtual environment and ChromaDB vector store. These are deployment prerequisites that affect out-of-the-box reproducibility, though they are not a limitation of the pipeline itself and are fully documented in the project README.md.

Section 3: Results and Discussion

3.1 Technical results:

The ingestion pipeline processed a total of 60 PDF documents — 5 PNRR funding calls and 55 EU Horizon WIDERA calls. Chunking these documents produced 509 text chunks in total, which were successfully stored in the ChromaDB vector database. A validation step confirmed that the number of chunks stored matched the number generated, ensuring full ingestion integrity.

PNRR Ranking Dominance: Limitation

Due to the fact that we were unable to find the entire EU Horizon WIDERA calls in their entirety, we have used the call summaries which we have found on the official European Commission website. This causes a pattern that emerges across all six of our universities strategic plans: every top-3 recommendation consists exclusively of PNRR calls, with no EU Horizon WIDERA call appearing in any top-3. This is likely explained by the difference in document length between PNRR calls (20+ pages) and WIDERA summaries (1–3 pages), which results in richer retrieval signal for PNRR documents.

Each strategic plan was then used to query the vector store, and the top-3 most relevant funding calls per university were identified through a composite scoring system comprising three components: a semantic score, a coverage score, and a consistency score. The ranking results show that Federico II and Politecnico di Milano achieved the highest average final scores (0.8250 and 0.8169, respectively) across their top-3 recommendations, while Bocconi and LUISS achieved the lowest scores (0.7730 and 0.8005 respectively). This outcome highlights a performance gap between strategic plans from public and private institutions, with public universities consistently outperforming their private counterparts.

Phase 2 introduces a deterministic XAI (Explainable AI) layer on top of the ranked results, producing structured evidence for each recommendation that shows which themes are strongly matched between the SP and the funding call, and where the SP underrepresents themes the call prioritises. Analysis of the XAI theme signals across all six universities reveals that strategic plans are most heavily oriented towards AI, Data & Digital Transformation (31.6%) and Internationalization & Collaboration (28.1%). The recommended funding calls, by contrast, emphasise Innovation Transfer & Industry (31.6%) and AI & Digital themes (29.5%). This reveals a consistent structural gap: while universities prioritise academic themes such as internationalization and digital transformation in their strategic plans, the highest-scoring funding calls disproportionately reward industry engagement and technology transfer – areas where SP coverage across all six institutions averages only 14.0%, less than half the signal found in the calls themselves. This misalignment is precisely what the XAI gap detection layer is designed to surface and flag for each institution.

In Phase 3, a locally-run Qwen3:0.6b language model generates a structured stakeholder briefing for each strategic plan, grounded strictly in the evidence provided. A citation policy is enforced automatically: each recommended call must include at least two explicit source citations (file name and page number). If the model failed to include citations present in the supporting evidence, they are injected from the retrieved chunks, and if a call still falls short, its confidence rating is downgraded to Low, preventing unsupported claims from reaching the output without signal.

3.2 Project results and business value

The observed score gap between public and private universities reflects a structural difference in how strategic plans are written. Public universities, being more reliant on public grant funding, tend to produce strategic plans whose language and priorities align naturally with public funding call criteria. Private institutions, which can draw on alternative revenue streams, orient their strategic documentation more toward institutional positioning and market differentiation. This suggests the tool is most immediately actionable for public universities, while private institutions may benefit from guidance on reorienting strategic plan language before applying.

The deliverable presented to EY is a web application implementing the full three-phase pipeline through an interactive interface. The interface is designed for immediate institutional use: funding administrators can explore pre-computed results for the six demo universities instantly, or upload any new strategic plan to receive tailored recommendations in real time. The ability to adjust pipeline parameters directly within the interface means that analysts can examine how recommendations shift under different retrieval thresholds and scoring weights, supporting more informed and defensible funding application decisions. Beyond the current scope, the interface is designed for generalisability, meaning any university can upload its strategic plan in any language supported by the embedding model. The corpus is currently scoped to European funding instruments (PNRR and Horizon Europe), but the architecture is modular and could be extended to additional funding landscapes, such as national schemes or non-European programmes, with a broader document corpus.

The XAI layer is central to the product's value proposition. Rather than producing opaque, black-box rankings, every recommendation is accompanied by matched themes, supporting evidence excerpts, and identified gaps, making the output fully auditable by stakeholders who need to justify or act on the suggestions. This level of transparency is particularly important in an institutional context, where decisions around funding applications involve multiple parties across administrative, research, and strategic functions.

Unlike conventional semantic search tools that return ranked results without justification, this system produces fully auditable, evidence-grounded recommendations operable entirely on local hardware – requiring no external API calls and exposing no sensitive institutional documents to third-party services. This combination of transparency, traceability, and privacy preservation represents the core differentiating element of the proposed solution.
The Phase 3 LLM output further lowers the barrier to adoption. By generating a plain-language executive briefing before presenting the technical detail, the system allows administrative staff and non-technical decision-makers to engage with the recommendations without needing to interpret raw scores or JSON outputs.

Section 4: Conclusions

4.1 Key takeaways:
This project demonstrates that a three-phase RAG pipeline combining dense semantic retrieval, deterministic explainability, and locally-run language generation can produce funding call recommendations that are both technically sound and immediately usable by non-technical stakeholders. The system successfully processed six Italian university strategic plans against a corpus of 60 funding call PDFs, returning ranked, evidence-backed, and confidence-graded recommendations for each institution.

A central finding is that explainability is not merely a desirable feature but a functional requirement in this domain. Pure semantic retrieval (Phase 1 alone) identifies relevant calls but offers no justification, making it unsuitable for institutional decision-making. The addition of the XAI layer (Phase 2) and the LLM briefing (Phase 3) transforms a ranking score into an auditable, actionable output that grant officers and administrators can act on directly.

The choice to run the pipeline entirely locally, using Qwen3-Embedding-0.6B for embeddings and Qwen3:0.6b for generation, demonstrates that a privacy-preserving, cost-free deployment is viable without relying on external APIs, which is a meaningful practical consideration for institutions handling sensitive strategic documents.

4.2 Natural next steps for this project

Several directions for future work follow naturally from the current limitations of the system, many of which stem from the computational constraints inherent to an academic project.
First, the corpus is presently scoped to European funding instruments, PNRR and Horizon Europe, evaluated against six Italian universities. This scope reflects the data availability and time constraints of the project rather than an architectural limitation. With greater resources, the funding call corpus could be extended to include national schemes from other countries and instruments relevant to non-European institutions, allowing the system to serve universities globally.

Second, the pipeline relies on Qwen3:0.6b, a compact model chosen specifically for its local deployability given the limited computational power available for this project. While it performs adequately for structured output generation, a larger model would likely produce richer and more fluent Phase 3 stakeholder briefings, support longer and more nuanced outputs, and reduce the occasional repetition that can occur with smaller architectures. Scaling to a more capable model is the most direct path to improving output quality without changing the pipeline architecture.

Finally, because no ground-truth benchmark dataset of correct university–call pairings exists, evaluation remains qualitative. A natural next step would be to construct such a benchmark with input from domain experts such as university grant offices, enabling quantitative measurement of ranking quality through standard information retrieval metrics.

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Appendix A:

```
INPUT: funding call PDFs, university strategic plan PDFs

// Phase 0 - Ingest
FOR each funding call PDF:
    extract and clean text → chunk (1800 chars, overlap 250)
    embed chunks with Qwen3-Embedding-0.6B
    store vectors + metadata in ChromaDB

// Phase 1 - Rank
extract and clean SP text → chunk (1200 chars)
FOR each SP chunk:
    query ChromaDB → retrieve top-30 similar call chunks
    aggregate scores per call:
        semantic_score, coverage_score, consistency_score
        final_score = 0.7*semantic + 0.2*coverage + 0.1*consistency
RETURN top-3 ranked calls

// Phase 2 - XAI
FOR each top call:
    compute theme signal on SP text and call evidence
    derive matched themes and key gaps via lexicon thresholds

// Phase 3 - LLM
build JSON payload from Phase 2 results
prompt Qwen3:0.6b → parse JSON + text response
validate and patch citations → downgrade confidence if needed
RETURN stakeholder briefing
```

Algorithm 1. *High-level pipeline pseudocode*

Algorithm 1 provides a high-level overview of the three-stage pipeline architecture, from data collection and vector store construction through to ranked recommendations, XAI scoring, and final LLM briefing generation.

Appendix B:

Who did what– CRediT:

Luka Burmazevic– Methodology, Conceptualization, Software, Visualization, Resources, Investigation

Evelina Ristovska– Methodology, Conceptualization, Software, Data Curation, Validation, Writing - Original Draft

Filippa Grönberg– Methodology, Conceptualization, Software, Data Curation, Writing - Original Draft and Review & Editing

In this work, we employed two main Generative AI tools: Anthropic Claude (Sonnet 4.6) and OpenAI Codex (GPT-5.3-Codex). Claude was used during the brainstorming and ideation phases to explore design alternatives for the retrieval and explainability pipeline, and to support editing portions of the technical report. OpenAI Codex were used throughout the development of the source code, including pipeline_core_methods.py, llm_explainer.py, and the ingestion and query notebooks, to assist with code completion, debugging, and refactoring. All AI-generated content, including code and text, was reviewed, validated, and tested by the authors prior to submission. Full responsibility for the submitted materials remains with the project team.